RAJIA RANI

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SUMMARY

MS in AI and Research Assistant at the University of South Dakota, specializing in LLM pre-training, distributed systems, and GPU performance optimization. 2.5 years of hands-on experience building and optimizing AI systems across startup and academic environments.

TECHNICAL SKILLS

Programming & Cloud:	Python, Java, Google Cloud (GCP), AWS, Docker, GPU Computing (CUDA)
Pre-Training & Inference:	Tensor Parallelism, Pipeline Parallelism, Sequence & Context Parallelism, ZeRO (Stage 1/2/FSDP), Gradient Accumulation, Ring-All Reduce, Activation Recomputation, Memory Optimization
Machine Learning & GenAI:	ML, DL, Transformers, Attention Mechanisms, Scaling Laws, LLMs, Instruction Tuning, LoRA/QLoRA, Diffusion Models, GPU Acceleration, Distributed Training
Agentic AI & LLM Systems:	LangGraph, FastAPI, MCP Tool Contracts, ChromaDB, Vector Retrieval, RAG Pipelines, Agent Orchestration, OpenTelemetry
NLP & Vision:	NLP, Tokenization (BPE, WordPiece), SentencePiece, Vector Search, Computer Vision, Vision-Language Models (VLMs), Multimodal Architectures
Math & CS Foundations:	Statistics, Linear Algebra, Probability, Optimization, Calculus, Data Structures & Algorithms
Data & Tools:	NumPy, Pandas, MongoDB, MySQL, Git, GitHub, Flask, Node.js, REST APIs

PROFESSIONAL EXPERIENCE

Research Assistant (RA) - University of South Dakota, USA **Jan 2026 – Present**

- Implemented an LLM pre-training pipeline using Megatron-DeepSpeed with 5D parallelism (DP+TP+PP+CP/SP+EP) and ZeRO (stage 2/3) across 8 GPUs; improved tokens/sec by 28% while maintaining 85% scaling efficiency.
- Profiled end-to-end pre-training step time & systematically tuned micro-batch size, gradient accumulation steps, communication overlap strategies; reduced step time from 520ms to 410ms and increased overall GPU utilization from 62% to 81%.
- Enabled long-context training via sequence/context parallelism and activation recomputation; extended sequence length from 2K to 8K tokens with 35% lower peak memory and 18% throughput improvement.

AI & Full-Stack Developer | Co-Founder - Majorana AI Labs Pvt. Ltd., India **Oct 2023 – Dec 2025**

- Built and deployed end-to-end AI-driven applications integrating machine learning models into production workflows; developed backend services in Python and Node.js, designed RESTful APIs, and managed scalable cloud infrastructure on AWS and GCP.
- Co-founded Majorana AI Labs and led development of AI-based solutions in computational chemistry, molecular modeling, drug discovery, and automation; drove product design, system architecture, and cross-functional research across scientific domains.

Chemistry Instructor - St. Soldier College, India **July 2019 – July 2023**

- Taught Organic Chemistry to undergraduate & senior secondary students, emphasizing conceptual clarity & practical application.
- Organized and coordinated seminars, workshops, and conferences to enhance student engagement and research exposure.

PERSONAL PROJECTS

Scalable LLM Training with 5D Parallelism *Github*

- Engineered a distributed GPT-2 training framework across 8× H100/A100 GPUs implementing all 5 parallelism strategies (DP, TP, PP, CP, EP) from scratch using PyTorch Distributed and NCCL, achieving ~1.5M tok/sec on DP=8 baseline.
- Using fine-tuning LORA.
- Built Tensor Parallelism (Column/Row linear layers), Pipeline Parallelism (1F1B scheduling), Ring Attention for long-context modeling, and MoE with top-k routing and all-to-all dispatch, enabling memory-efficient training across 7 parallel configurations.
- Benchmarked throughput and memory tradeoffs across configurations (DP to 5D), identifying that full 5D composition (TP=2, PP=2, EP=2) reduced per-GPU memory footprint while maintaining convergence (loss ~ 5.49).

PulseGuard AIOps - AI Incident Commander *Github*

- Built a 10-node LangGraph agent workflow for automated incident triage, orchestrating evidence collection across service health, deployments, feature flags, dependency health, and service metadata through 5 deterministic MCP-style tool contracts.
- Developed a production-style FastAPI backend with ChromaDB vector retrieval over runbooks and postmortems for semantic lookup; built an offline evaluation harness validating severity classification, root-cause prediction, tool usage, & retrieval quality.

RAG-Based Information Retrieval System *Github*

- Designed a RAG pipeline from scratch without LangChain/LlamaIndex, implementing fixed, semantic, and structural chunking with hybrid BM25, IVF indexing, and ms-marco cross-encoder reranking on AWS.
- Integrated FAISS, Chroma, Supabase (pgvector) for cosine similarity search; evaluated with RAGAS, delivering citation-backed responses with source text and page attribution.

EDUCATION

MS in Artificial Intelligence - *University of South Dakota, USA* **01/2026 - Present**

Master of Science (Chemistry) - *Lovely Professional University, India* **08/2017 - 06/2019**

Bachelor of Science (Math, Physics, Chemistry) - *PCM. S.D. College for Women, India* **06/2014 - 05/2017**